

VoiceTranslate Sequence Flow

Systems involved:

- User
- Browser / React Frontend
- Microphone or File Upload
- FastAPI Backend
- Local Temporary Storage
- Guardrail Service
- OpenAI Speech-to-Text
- Audio Analysis Service
- OpenAI Translation LLM
- ElevenLabs Text-to-Speech
- Media Endpoint

Sequence:

1. User opens the VoiceTranslate app in the browser.
2. React frontend calls FastAPI: GET /api/health.
3. User grants microphone permission or uploads an audio file.
4. Browser captures English speech as an audio blob.
5. User confirms voice consent and submits the request.
6. Frontend sends multipart form data to POST /api/translate/voice.
7. FastAPI creates a request ID and validates consent.
8. Backend saves the uploaded audio in temporary local storage.
9. Audio validation checks file format and size.
10. OpenAI STT transcribes the English audio.
11. Guardrails check content safety and impersonation risk.
12. Audio analysis extracts pitch, energy, pace, pauses, and mood.
13. OpenAI LLM translates the transcript into natural spoken Hindi.
14. Guardrails check PII and collect warnings.
15. ElevenLabs generates Hindi speech using the configured voice ID.
16. Backend saves the generated audio temporarily.
17. Backend returns JSON with transcript, translation, mood profile, guardrails, and audio URL.
18. Browser loads the generated audio through the media endpoint.
19. User plays or downloads the translated Hindi audio.

Compact sequence diagram:

User -> Browser: Open app
Browser -> FastAPI: GET /api/health
User -> Browser: Grant mic / upload audio
Browser -> FastAPI: POST /api/translate/voice
FastAPI -> Guardrails: Consent check
FastAPI -> Storage: Save uploaded audio
FastAPI -> OpenAI STT: Transcribe audio
FastAPI -> Guardrails: Safety + impersonation checks
FastAPI -> Audio Analysis: Detect mood and voice features
FastAPI -> OpenAI LLM: Translate English to Hindi

FastAPI -> ElevenLabs: Generate Hindi cloned voice audio

FastAPI -> Storage: Save generated MP3

FastAPI -> Browser: Return result + audio URL

Browser -> Media Endpoint: Load audio

User -> Browser: Play or download audio